

Student grade system

Describing the task:

The Student Grade Management System is a menu-driven C++ application developed using Object-Oriented Programming (OOP) concepts. The system is designed to manage student records efficiently by allowing users to add, view, search, update, and delete student information. It also calculates each student's total marks, percentage, and grade based on marks entered for multiple subjects.

The project uses classes and inheritance to organize data, STL containers (vector) to store student records dynamically, and file handling to save and load records from a text file, ensuring data persistence between program executions. The application provides a user-friendly menu interface for performing all operations and includes basic unit tests to verify the correctness of important functions.

Compile command:

```
g++ GradeSystem.cpp main.cpp person.cpp student.cpp
```

Run command:

```
./a.exe
```

Output Screenshots:

=====

STUDENT GRADE MANAGEMENT SYSTEM

=====

1. Add Student
2. Display All Students
3. Search Student
4. Update Student
5. Delete Student
6. Sort By Name
7. Sort By Percentage
8. Save Records
9. Load Records
10. Student Count
0. Exit

=====

Enter your choice: 0

=====

Thank You!

Exiting Student Grade Management System...

=====

=====

Enter your choice: 8

101

Rahul

20

100.00 95.00 90.00 85.00 80.00

Records saved successfully.

=====

STUDENT GRADE MANAGEMENT SYSTEM

=====

1. Add Student
2. Display All Students
3. Search Student
4. Update Student
5. Delete Student
6. Sort By Name
7. Sort By Percentage
8. Save Records
9. Load Records
10. Student Count
0. Exit

=====

Enter your choice: 10

Total Students = 1

Enter your choice: 3

Enter Student ID to search: 101

Student Found

```
-----
Student ID : 101
Name       : Rahul
Age        : 20
Marks      : 100.00 95.00 90.00 85.00 80.00
Total      : 450.00
Percentage : 90.00%
Grade      : A
-----
```

```
=====
          STUDENT GRADE MANAGEMENT SYSTEM
=====
```

1. Add Student
2. Display All Students
3. Search Student
4. Update Student
5. Delete Student
6. Sort By Name
7. Sort By Percentage
8. Save Records

```
=====
Student 1
```

```
-----
Student ID : 101
Name       : Rahul
Age        : 20
Marks      : 100 95 90 85 80
Total      : 450
Percentage : 90.00%
Grade      : A
-----
```

Total Students : 1

```
=====
          STUDENT GRADE MANAGEMENT SYSTEM
=====
```

1. Add Student
2. Display All Students
3. Search Student
4. Update Student
5. Delete Student
6. Sort By Name
7. Sort By Percentage
8. Save Records
9. Load Records

```
Enter Marks of 5 Subjects
Subject 1 : 100 95 90 85 80
Subject 2 : Subject 3 : Subject 4 : Subject 5 :
Student added successfully.
```

```
=====
          STUDENT GRADE MANAGEMENT SYSTEM
=====
```

1. Add Student
2. Display All Students
3. Search Student
4. Update Student
5. Delete Student
6. Sort By Name
7. Sort By Percentage
8. Save Records
9. Load Records
10. Student Count
0. Exit

```
=====
Enter your choice: 2
```

```
=====
          STUDENT RECORDS
=====
```

```
PS C:\Users\DELL\html\.vscode\intership_tasks\student grade system> g++ GradeSystem.cpp main.cpp person.cpp student.cpp
PS C:\Users\DELL\html\.vscode\intership_tasks\student grade system> ./a.exe
```

```
=====
          STUDENT GRADE MANAGEMENT SYSTEM
=====
```

1. Add Student
2. Display All Students
3. Search Student
4. Update Student
5. Delete Student
6. Sort By Name
7. Sort By Percentage
8. Save Records
9. Load Records
10. Student Count
0. Exit

```
=====
Enter your choice: 1
```

Enter Student Details

Enter Student ID : 101

Enter Student Name : Rahul

Enter Age : 20